

Contact

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www.linkedin.com/in/adiyamanglik
(LinkedIn)

adiyamanglik.github.io/ (Personal)

Top Skills

Entrepreneurship

Public Speaking

Computer Architecture

Languages

English (Native or Bilingual)

Hindi (Native or Bilingual)

Punjabi (Professional Working)

German (Professional Working)

Certifications

Keynote Speaker

Green Software for Practitioner

Speaker: Open Source Summit
North America 2024

Nand2tetris

Speaker: Embedded Open Source
Summit 2024

Honors-Awards

First Position, Microsoft Azure
Machine Learning Hackathon

Second Position, APOGEE 2016
Project Exhibition

Undergraduate research scholar

Scholarship for International
Exchange studies

High-Performance Computing school

Publications

Resistive Crossbar-Aware Neural
Network Design and Optimization

NEON: Enabling Efficient Support
for Nonlinear Operations in Resistive

Aditya Manglik

Senior Engineer

Cork, County Cork, Ireland

Summary

Five+ years of industry experience, complemented by specialized research at Columbia University and ETH Zurich. My work focuses on building high-performance, secure, and energy-efficient computing systems. I have a proven track record of architecting and optimizing systems from the microarchitecture level up to distributed cloud services, and my research has been published in top-tier research conferences.

Experience

Qualcomm

Senior Engineer

July 2025 - Present (11 months)

Cork, County Cork, Ireland

- Technologies: AI Hardware design, Transformers, Deep Learning, C++, SystemC, Python, SoC design, NoC, Computer Architecture, DRAM, HBM, SOC performance, SOC architecture, machine learning

- Optimized memory controller design in System-on-Chip (SoC) to maximize DRAM performance and energy efficiency

- Focus on DRAM Technologies: DDR5, DDR4, LPDDR6, LPDDR5, HBM

- Focus on driving innovation via patents and research in Processing-in-Memory (PIM)

ETH Student Project House

Founder

January 2025 - June 2025 (6 months)

Zurich, Switzerland

- Attempted commercializing my ETH Zurich research focusing on private and sustainable LLM services

- Built benchmarking infrastructure for GPU kernel-level profiling and end-to-end SLOs (latency/throughput/energy)

- Conducted customer discovery interviews; narrowed product market fit and validated buying criteria (cost, privacy, latency)

- Consulted for other AI start-ups, gained expertise in achieving SOC 2 and ISO 27001 compliance
- Wound down due to administrative and bureaucratic challenges

Augmedi

Software Engineer

June 2024 - December 2024 (7 months)

Zurich, Switzerland

Technologies: TypeScript, React, SQL, Three.js, DevOps, Tanka, Terraform, Kubernetes

- Built and shipped features using TypeScript, React, and SQL.
- Designed and implemented end-to-end DevOps features, improving platform performance and maintainability.
- DevOps: Delivered Grafana and Prometheus monitoring dashboards, fixed service outages (on-call midnight).
- Direct interaction with early customers and adopters, strong experience in translating pain points into features and scalable solutions.

Columbia University

Research Staff

September 2023 - April 2024 (8 months)

New York, United States

Advisor: Prof. Kostis Kaffes (Columbia University in the City of New York), Prof. Lana Josipović (ETH Zürich)

Technologies: Golang, Bash, Python, Serverless, Performance Engineering

Focus: Optimizing the performance of garbage collection in Java, Go, and JavaScript run-times for serverless workloads

- * Implemented garbage collection-aware request scheduling, reduced latency deviation by 2.5x.
- * Conducted extensive benchmarking using Python and Golang to validate load balancer and scheduling algorithms.

ETH Zürich

Graduate Research Assistant

September 2021 - August 2023 (2 years)

Zurich, Switzerland

Advisor: Prof. Onur Mutlu (<https://people.inf.ethz.ch/omutlu/>).

- * Conducted research in the SAFARI Research Group with Prof. Onur Mutlu

* Specialized in hardware acceleration for deep learning, Processing-in-Memory (PIM), and DRAM systems.

* Additional study in compilers, Instruction Set Architecture, microarchitecture, and hardware security vulnerabilities including Rowhammer, Meltdown, and Spectre

* List of Publications:

* * Ivan Fernandez, Aditya Manglik, Christina Giannoula, Ricardo Quisilant, Nika Mansouri Ghiasi, Juan Gómez-Luna, Eladio Gutierrez, Oscar Plata, Onur Mutlu (2024). MATSA: An MRAM-Based Energy-Efficient Accelerator for Time Series Analysis. IEEE Access.

* * Aditya Manglik, Minesh Patel, Haiyu Mao, Behzad Salami, Jisung Park, Lois Orosa, Onur Mutlu (2022). NEON: Enabling Efficient Support for Nonlinear Operations in Resistive RAM-based Neural Network Accelerators. Arxiv.

* * Nika Mansouri Ghiasi, Mohammad Sadrosadati, Geraldo F Oliveira, Konstantinos Kanellopoulos, Rachata Ausavarungnirun, Juan Gómez Luna, Aditya Manglik, João Ferreira, Jeremie S Kim, Christina Giannoula, Nandita Vijaykumar, Jisung Park, Onur Mutlu (2022). RevaMp3D: Architecting the Processor Core and Cache Hierarchy for Systems with Monolithically-Integrated Logic and Memory. Arxiv.

* * Haiyu Mao, Mohammed Alser, Mohammad Sadrosadati, Can Firtina, Akanksha Baranwal, Damla Senol Cali, Aditya Manglik, Nour Almadhoun Alserr, Onur Mutlu (2022). GenPIP: In-Memory Acceleration of Genome Analysis via Tight Integration of Basecalling and Read Mapping. MICRO-55.

* * Minesh Patel, Taha Shahroodi, Aditya Manglik, A Giray Yaglikci, Ataberk Olgun, Haocong Luo, Onur Mutlu (2022). A case for transparent reliability in DRAM systems. Arxiv.

Oracle

Cloud Engineer

August 2018 - September 2021 (3 years 2 months)

Bengaluru Area, India

Technologies: Java, REST, API, Python, Semantic Classification, Performance Engineering, JVM, Code review, Git, Unit test

* Application feature development using Java, JUnit, and J2EE platforms

- * Optimized J2EE component latencies to improve cloud-native CRM application's response time from 450 to 300 ms (33% improvement).
- * Hosted internal brown bag sessions on machine learning as an early adopter

GNOME Foundation

Google Summer of Code

May 2018 - September 2018 (5 months)

Austria

News and media coverage:

1. https://www.phoronix.com/scan.php?page=news_item&px=GNOME-Usage-Power-2018
2. <https://adrienplazas.com/blog/2022/07/25/berlin-mini-guadec-2022.html>
3. <https://www.phoronix.com/news/GNOME-Usage-Power-2020>
4. <https://lfenergy.org/lf-energy-summit-2023-recap-understanding-and-measuring-the-carbon-footprint-of-personal-computing/>

Link: <https://summerofcode.withgoogle.com/archive/2018/projects/5742960490577920/>

Technologies: C, Linux kernel, Regression, Data collection and cleaning, GTK, Vala, Energy, Power, ACPI, Battery

- Developed a Linux kernel module to quantify a process' energy consumption over its lifetime
- Used multi-variate regression to quantify the power consumption of individual hardware devices from battery drain rate

Technische Universität Wien

Research Staff

January 2018 - June 2018 (6 months)

Austria

Advisor: Prof. Muhammad Shafique (<https://ti.tuwien.ac.at/ecs/people/shafique>)

Research in deep learning, convolutional neural networks, transformers, Processing-in-Memory, Memristor, Resistive RAM (ReRAM), Pruning, Ensembles, MoE, Neuromorphic Computing, In-Memory Computing

Research paper:

Aditya Manglik, Muhammad Hanif Abdullah, Muhammad Shafique (2021).
Resistive Crossbar-Aware Neural Network Design and Optimization (IEEE
Access)

CSIR-CEERI

Embedded System Engineer
May 2016 - July 2016 (3 months)
Chennai, Tamil Nadu

Embedded programming of ARM Cortex-M7 based STM32F746
microcontroller with STM32F4 camera

Experience with Arduino, Beaglebone Black, and Raspberry Pi programming
and system development

Education

ETH Zürich

Master of Science - MS, Electrical Engineering · (2021 - 2024)

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering - BE, Electrical and Electronics Engineering (with
Honors) · (2014 - 2018)

Delhi Public School - R. K. Puram